

DR. MOHAMMED MOUZAKKIR HOSSAIN

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Research gate



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Permanent / Present Address

Vill : Chamtibagan, P.O. : Tejhati, P.S. : Nalhati, Dist. : Birbhum State : West Bengal, Pin Code : 731220,

Country : India.

Career profile

- Highly motivated researcher with five publications in reputed journals in 4 years.
- Ability to work in a multi-cultural workspace with expertise in cross-cultural communication skills, forward-thinking, and decision-making.
- Curiosity and task-driven with consistent documentation and analysis of numerical output, which results in the successful completion of projects over the due time.

Areas of Teaching Interest

ODEs & PDEs, Numerical Analysis, Fluid dynamics, Complex Analysis, Calculus, Abstract Algebra, Linear Algebra.

Areas of Research Interest

Computational fluid dynamics, Hydrodynamics stability analysis, Numerical and mathematical methods.

Research Job

- Ongoing** ■ Postdoctorate (**School of Energy Science and Engineering**), Harbin Institute of Technology, China.
Research Area: *Computational Fluid Dynamics*

Education

- Feb, 2024 ■ Ph.D. (**Mathematics**), SRM Institute of Science and Technology, Tamil Nadu, India.
Research Area: *Computational Fluid Dynamics (Specifically, hydrodynamic stability analysis)*
Thesis title: *Dynamics and Instability Mechanisms of single and Double-layered Fluid flow Models.*
Duration: 4.2 years.
- June, 2016 ■ M.Sc. (**Applied Mathematics**), University of Calcutta, West Bengal, India.
- June, 2013 ■ B.Sc. (**Mathematics**), University of Burdwan, West Bengal, India.

Education (continued)

- May, 2009  **H.S. (English, Bengali, Physics, Chemistry, Mathematics)**, WBCHSE, West Bengal, India.
- 2007  **M.P., (English, Bengali, Mathematics, History, Geography, Physical Science, Life Science)**, WBBSE, West Bengal, India.

National / State Level Examinations

- 2017  **WBSET** (West Bengal State Eligibility Test), conducted by the West Bengal College Service Commission, Govt. of West Bengal state.
- June 2018  **CSIR-UGC NET** (National Eligibility Test), conducted jointly by the University Grant Commission and Council of Scientific and Industrial Research, Govt. of India.
- Dec 2018  **CSIR-UGC NET** (National Eligibility Test), conducted jointly by the University Grant Commission and Council of Scientific and Industrial Research, Govt. of India.
- 2018  **GATE** (General Aptitude Test in Engineering), conducted by the National Coordination Board, Dept. of Higher Education, Govt. of India.
- 2021  **GATE** (General Aptitude Test in Engineering), conducted by the National Coordination Board, Dept. of Higher Education, Govt. of India.

Achievements

- 2019–2022  **Junior Research Fellow**
DST-SERB sponsored project, SRM Institute of Technology, Project Titled “SERB, Department of Science and Technology, Government of India through CRG project, Award No. CRG/2018/004521, .
- 2022  **Senior Research Fellow**
DST-SERB sponsored project, SRM Institute of Technology, Project Titled “SERB, Department of Science and Technology, Government of India through CRG project, Award No. CRG/2018/004521.
- 2022 – 2024  **Temporary Junior Research Fellow**
DST-SERB sponsored project, Indian Institute of Technology Jodhpur, Project Titled “Mathematical framework of secondary and nonlinear instabilities for complex flow systems” Project no: S/SERB/SKG/20210097.

Research Publications (Published/Accepted)

Journal Articles

-  1 D. Paul, **M.M. Hossain**, and H. Behera. “Hydrodynamic instability of odd-viscosity-induced shear-imposed falling film”. In: *Physics of Fluids, American Institute of Physics* 35.3 (2023).

- 2 D. Paul, **M.M. Hossain**, and H. Behera. "Hydrodynamic stability analysis of shear-layered fluid flow over a porous bed in the presence of a floating elastic plate". In: *International Journal of Non-Linear Mechanics, Elsevier* 159 (2024), p. 104599.
- 3 **M.M. Hossain**, C.C. Tsai, and H. Behera. "Instability mechanism of shear-layered fluid in the presence of a floating elastic plate". In: *Physics of Fluids, American Institute of Physics* 35.2 (2023).
- 4 **M.M. Hossain** and H. Behera. "Shear-imposed falling thin Newtonian film over a porous slippery surface". In: *Physics of Fluids, American Institute of Physics* 34.11 (2022).
- 5 **M.M. Hossain**, S. Ghosh, and H. Behera. "Impact of a floating flexible plate on the stability of double-layered falling flow". In: *Applied Mathematical Modelling, Elsevier* 132 (2024), pp. 454–474.
- 6 **M.M. Hossain**, S. Ghosh, and H. Behera. "Linear instability of a surfactant-laden shear imposed falling film over an inclined porous bed". In: *Physics of Fluids, American Institute of Physics* 34.8 (2022).
- 7 **M.M. Hossain**, S. Ghosh, and H. Behera. "Odd-viscosity induced surfactant-laden shear-imposed viscous film over a slippery incline". In: *Meccanica, Springer* (2024), pp. 1–21.
- 8 **M.M. Hossain et al.** "Hydrodynamic instability of shear imposed falling film over a uniformly heated inclined undulated substrate". In: *Physics of Fluids, American Institute of Physics* 36.8 (2024).
- 9 **M.M. Hossain et al.** "Stability dynamics of an odd-viscosity induced fluid flow over a vibrating inclined surface". In: *Physics of Fluids, American Institute of Physics* 37.10 (2025).

Technical Skills

Languages	■ Strong reading, writing, and speaking competencies in English, Hindi, and Bengali.
Programming	■ MATLAB
Technique	■ The idea of using numerical techniques (Chebyshev-spectral collocation method, finite difference method, etc.) and analytical techniques (Longwave expansion, Energy budget analysis, normal mode analysis) to study the dynamics and instability mechanism of different fluid flow configurations.
Databases	■ Strong in Productivity software like LaTeX, Microsoft Office Suite including Word, Excel, and PowerPoint.
Misc.	■ Experience in technical report writing and project presentations at meetings and publishing.

Miscellaneous Experience

International Conferences

- 2020 ■ MM. Hossain and H. Behera, "Surfactant-laden Newtonian falling film down a wavy channel: Linear and nonlinear stability analysis", International Conference on Advances in Differential Equations and Numerical Analysis (ADENA-2020), IIT Guwahati, India, 12th -15th October 2020.
- 2021 ■ MM. Hossain and H. Behera, "Hydrodynamic instability of a shear-imposed thin Newtonian gravity-driven film flow down a permeable inclined plane". 2nd International Conference on Mathematical Techniques and Applications (e-ICMTA-2021), SRM IST, 24th -26th March 2021.
- 2022 ■ MM. Hossain and H. Behera, "Linear instability of a surfactant-laden shear-imposed falling film over an inclined porous bed". 3rd International Conference on Mathematical Techniques and Applications (e-ICMTA-2022), SRM IST, 23rd -25th March 2022.

Miscellaneous Experience (continued)

Workshops/Seminars/Research Day

- MM. Hossain and H. Behera, “Effect of a floating flexible plate on the dynamics of two-layer film flow across an inclined permeable bed”. Research & Science Day, SRM IST, 1st March 2021.
- MM. Hossain and H. Behera, “Effect of a floating flexible plate on the dynamics of two-layer film flow across an inclined permeable bed”. Dr. Paarivendhar Research Colloquium (DPRC-2022), SRM IST, 24th -26th March 2022.
- International Workshop on Numerical and Analytical Techniques in Engineering Problems (IWNATEP-2022), SRM IST, 19th -21st January, 2022.
- International Workshop on Mathematical and Computational Methods in Science and Engineering (IWMCMSE-2023), IIT Ropar, 2nd -13th June, 2023.
- Symposium on Advanced Mathematical Modelling and Computing (SAMMC-2023), IIT Jodhpur, 5th -12th March, 2023.

References

Reference 1 : Dr. Harekrushna Behera

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Reference 2 : Dr. Sukhendu Ghosh

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Reference 3 : Prof. Chia-Cheng Tsai

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Reference 4 : Prof. Youchuang Chao

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Declaration

I hereby declare that all statements made here are true to the best of my knowledge and belief.

Md. Mouzakkir Hossain

Place: India

(Mohammed Mouzakkir Hossain)